



Society of Petroleum Engineers

SPE-230218-MS

Successful Remediation of Packer Integrity on a Newly Drilled Water Injection Well Suffering from Sustained Casing Pressure Utilizing Rigless Solution

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230218-MS](https://doi.org/10.2118/230218-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

A newly drilled water injection well encountered Sustained Casing pressure (SCP) during the commissioning phase. This had resulted in well to be kept in shut-in to avoid high Annulus A pressure posing serious HSE risk and further unable to meet the field water injection requirements.

A thorough SCP investigation was carried out which identified that the source of leak was the production packer which caused the SCP in Annulus A (i.e. space between Production tubing x Production casing). This paper will introduce the rig-less solution to restore the primary barrier integrity thus saving significant Work over cost & time and resuming the well injection successfully.

Whilst well was under injection mode, Annulus A pressure observed to be tracking with Well head injection pressure. SCP Investigation was carried out in accordance with API RP 90-2 guidelines. Annulus A Pressure was bled down to 0 psi but built back to the original pressure (before bled) within 24 hours despite the repeated bleed off attempts.

Production Packer suspected to be partially unset due to elastomer relaxation during water injection creating a leak path between Tubing to Production casing. Discussion was held with all stakeholders and agreed to re-energize the packer seal element by pressurizing the tubing up to 4500 psi using wireline plug and utilizing rig-less package.

After production packer re-energization operations, the injection was resumed in staged increments based on the injection rate requirements. Injection parameters were closely monitored for a week and Annulus A pressure observed to be stable at zero psi despite the wellhead injection pressure exceeding the requirements.

Introduction

ADNOC Onshore operates the "ABC" oil field, which was discovered in 1960s and producing from three major reservoirs namely X, Y and Z. Water Injector Wells are used to inject water into the formation to maintain reservoir pressure, thereby sustaining the oil production to achieve the production target. Currently there are around 300 water injection wells in the field and injecting at a rate of approximately 1 Million BWPd.

In field "ABC", water injection system consists of individual clusters. Each cluster consists of a water supply wells (WSW) equipped with an ESP, surface injection pump, chemical injection facility, power skid and surface piping connecting to the associated water injection wells of the cluster. The source of water is lifted up by ESP to surface facilities from Aquifers reservoirs.

Recently we have observed that newly drilled and completed water injection wells were experiencing with sustained casing pressure in the early stages of well life time post commissioning phase. This paper discusses about two well cases, where the SCP had developed at very early stage, process of trouble shooting involved and remedial activities carried out through rigless solution to restore the primary well barrier element i.e. packer, thereby preventing the HSE risk, saving the significant Work over cost and associated time and also fulfilling the water injection target requirements. Further, this paper explains about root cause investigation of packer element failure and the lessons learned to avoid similar issue in the future injection wells by implementing the best practices and guidelines.

Double Well Barrier Envelope Policy and SCP Investigation Process

Double well barrier envelope policy is a main philosophy of well integrity. It dictates all wells with positive pressure at surface and capable of flowing naturally should have primary and secondary barrier envelopes. Primary barrier envelope consists of well components which are continuously in direct contact with well fluid and exposed to pressure. Failure of any element of the primary barrier envelope results in the sustainable casing pressure in annulus "A" - SCP pressure. Such SCP wells are a violation of the double well barrier envelope policy. In the event of primary barrier failure e.g. leak through the tubing, packer, or other completion equipment, secondary barrier envelope must contain the pressure and prevent escalation of the incident outside of well envelope.

Main elements of the primary well barrier are tubing, completion equipment, tubing head seals, packer. The integrity of these components are evaluated upon the installation during drilling or workover phase, yet they can deteriorate over the operation period as a result of factors such as corrosion, erosion, mechanical wear, and fluctuations in pressure and temperature, as well as potential damage from well interventions.

The standard well barrier diagram provided below (Fig. 1) demonstrates this double well barrier envelope policy. The primary barrier is highlighted in red, while the secondary barrier is depicted in green.

Sustained Casing Pressure (SCP) is the unintended pressure in a contained annulus resulting from the flow of pressurized formation fluids in communication with the subject annulus that is measurable at the wellhead, rebuilds after having been bleed down and not contributed by well bore temperature fluctuations.

Confirmation of sustainable casing pressure of Annulus A should be carried out in accordance of API RP 90-2 or Company Well Integrity Management Strategy (WIMS) when well having Annulus A pressure more than 100 psi. The existing Annulus pressure should be bled 25-30 % below the original pressure (or maximum bled to zero psi) and the pressure build up need to be checked after 24 hours post bleed off. There will be 03 analysis of the bleed down and build up test

- **Pressure Bleeds Down without Build-up:** If the pressure bleeds to 0 psig and does not build up within 24 consecutive hours, the source of pressure in the annulus in question is either thermal in origin or results from a leak with a very low rate. In this case, the barriers for pressure containment can be considered effective.
- **Pressure Bleeds Down with Build-up:** If the pressure bleeds to 0 psig and builds back up to original or lower pressure within 24 consecutive hours, then a barrier associated with the annulus in question has a small leak.
- **Pressure Does Not Bleed Down:** If the pressure does not bleed to 0 psig within 24 hours, the pressure barrier may not be effective and, in some cases, the leak rate may be unacceptable. This condition can indicate that the leak rate is greater than the bleed rate.

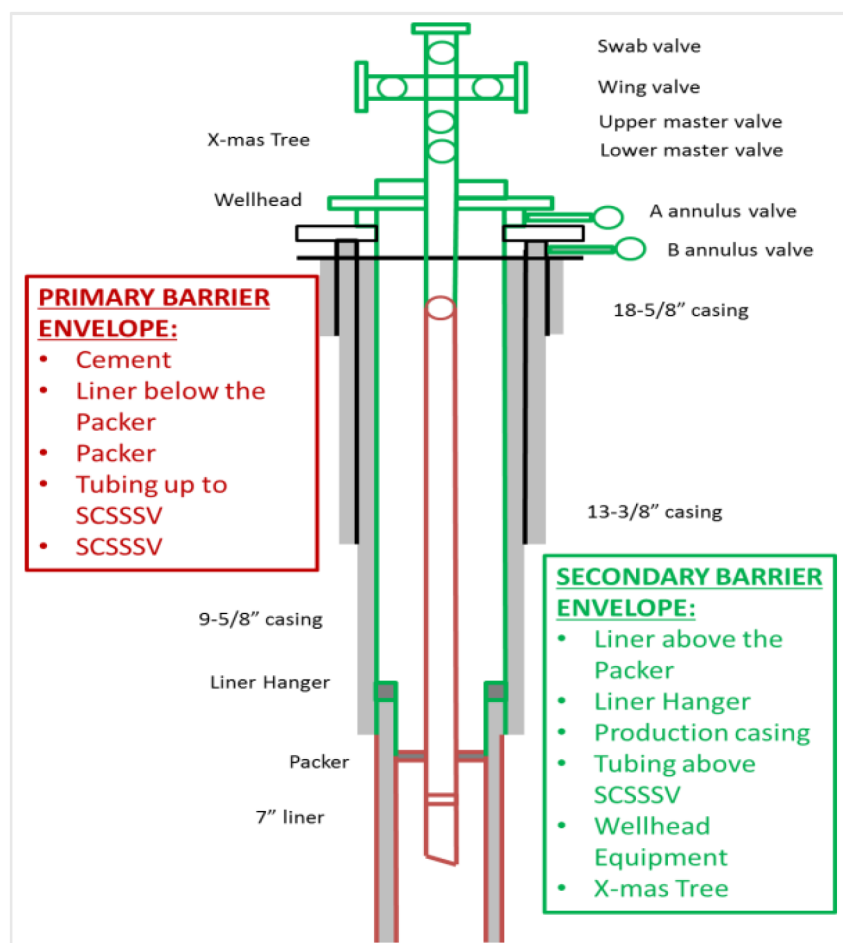


Figure 1—Typical well barrier schematic

SCP due to production packer leak can occur for several reasons, each affecting the integrity of the well and potentially leading to operational issues. Here are some common causes:

1. **Mechanical Failure:** The packer, designed to isolate sections of the wellbore, might fail due to mechanical issues such as wear and tear, manufacturing defects, or improper installation. Components like seals, slips, or hydraulics might malfunction.
2. **Corrosion and Erosion:** The harsh conditions within an oil well, including high pressures and temperatures, as well as the presence of corrosive fluids, can lead to corrosion and erosion of the packer materials. Over time, this can degrade the packer's ability to seal effectively.
3. **Chemical Reactions:** Certain chemicals used in drilling, completion, or production fluids can react with the materials in the packer, causing deterioration. For example, acids or other aggressive fluids can break down seals or other components.
4. **Pressure and Temperature Extremes:** Oil and gas wells often operate under extreme pressures and temperatures. If the packer is not rated for these conditions or if there are sudden changes in pressure or temperature, it can lead to leaks.
5. **Improper Energization:** If a packer is not properly set, it may not create a reliable seal. This could be due to incorrect energization procedures or equipment malfunctions.
6. **Mechanical Damage:** During drilling, completion, or workover operations, the packer may be subjected to mechanical damage from tools or equipment. This can compromise its ability to function correctly.

7. Operational Issues: Problems with well operations, such as excessive pressure or flow rates, can stress the packer and lead to leaks.
8. Environmental Factors: Changes in the surrounding geological environment, such as shifting formations or unexpected pressure changes, can impact the effectiveness of the packer.

The conventional process of curing SCP Annulus A due to production packer leak is to re-complete the well during rig-based workover. During this process, the old completion is pulled out of hole, and the new completion is installed. This process is repeated in every work over to restore the well integrity.

Production Packer Failure Cases

Case 1

Well "A" was drilled and completed in May'2021 as Single Water Injector (SWI) in "X" reservoir. Later in Feb'2024, as part of Development strategy, the well was worked over for horizontalization and re-completed as SWI with lower completion and equipped with production packer set in the liner. Well was reported with high Annulus A pressure of around 2000 psig in Annulus A during the commissioning phase. Bleed down attempts on Annulus A were unsuccessful as the pressure had re appeared in Annulus A and maximum of 2200 psi (or 60% of MAASP A) was noted under injection and pressure was re-building after 24-hour period (Refer Fig. 2 - Well "A" - Annulus "A" Bleed down and Build up test Record Before Packer Re-energization). A thorough SCP investigation was carried out by testing the tubing to 2400 psi confirming its integrity, which identified that the suspected source of SCP could be the production packer. Later Internal alignment meeting was held with all stake holders involving the OEM to define the way forward plan and investigate the root causes of premature failure of packer. It was suspected the packer unset or packer elastomer relaxation due to reduction of well bore temperature during water injection operation.

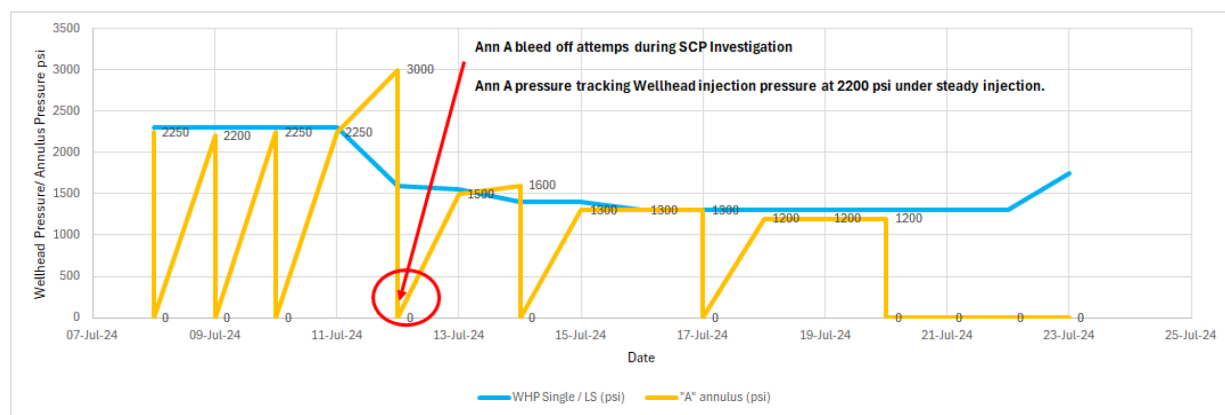


Figure 2—Well "A" - Annulus "A" Bleed down and Build up test Record Before Packer Re-energization

The team discussed the issue and agreed to attempt packer elastomer re-energization by pressurizing the tubing against N-test tool to 4500 psig/ 1 hour in stages. Tubing tested to 2000 psi/10 mins, 2500 psi/10 mins, 3000 psi/10 mins, 3500 psi/ 10 mins, 4000 psi/10 mins and 4500 psi/1 hour. The tubing pressure was holding at 4500 psi for 1 hour. During the test observed Annulus A pressure maximum climbed up to 400 psi due to ballooning effect. Upon successful test, the plug was retrieved and injection resumed at 2200 psi pressure in staged increment with 0 psi build up in Annulus A observed within 02 days. The magnitude of Annulus A pressure had decreased to min. of 300 psi (8.3 % of MAASP Annulus A) at 2200 psi injection pressure under steady injection mode which is significantly less compared with previous well injection condition i.e. before packer elastomer re-energization operations (Refer Fig. 3 - Well "A" - Annulus "A" Bleed down build up test post 1st Cycle Packer re-energization). Further discussion held with the team on

the well behavior and agreed to attempt the 2nd cycle of Packer re-energization by repeating the same steps. Upon completion of 2nd cycle, well resumed injection with 2200 - 2400 psi injection pressure and continued observing well pressure behavior on daily basis. Later, it was observed Annulus A noted to be around min 90 psi (2.5 % of MAASP Annulus A) at 1500 psi injection pressure under steady injection mode (Refer Fig. 4 - Well "A" - Annulus "A" Bleed down build up test post 2nd Cycle Packer re-energization). Later the well "A" was allowed to continue to operate under dispensation ensuring the Annulus A pressure must not exceed 50 % of MAASP of Annulus A (or 1800 psi).

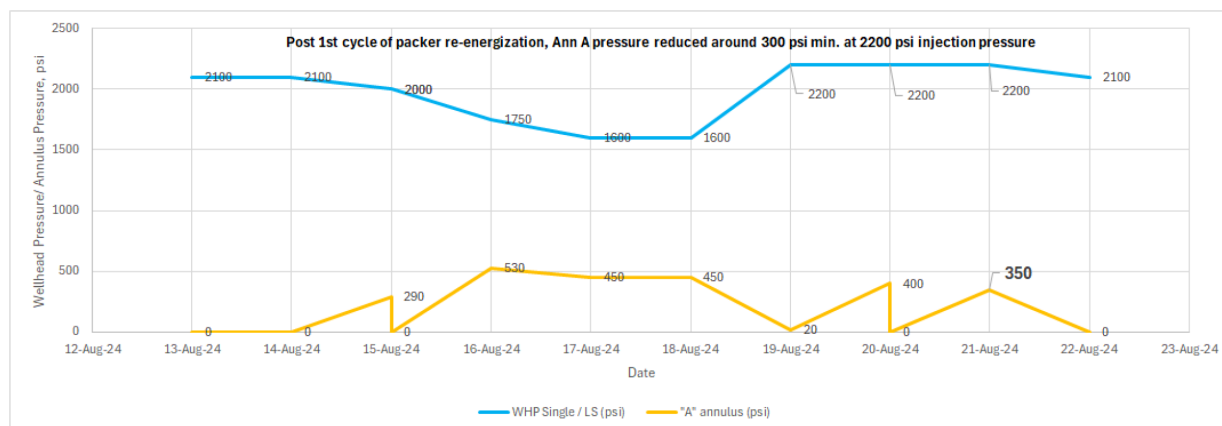


Figure 3—Well "A" - Annulus "A" Bleed down and Build up test Record Post 1st Cycle Packer Re-energization

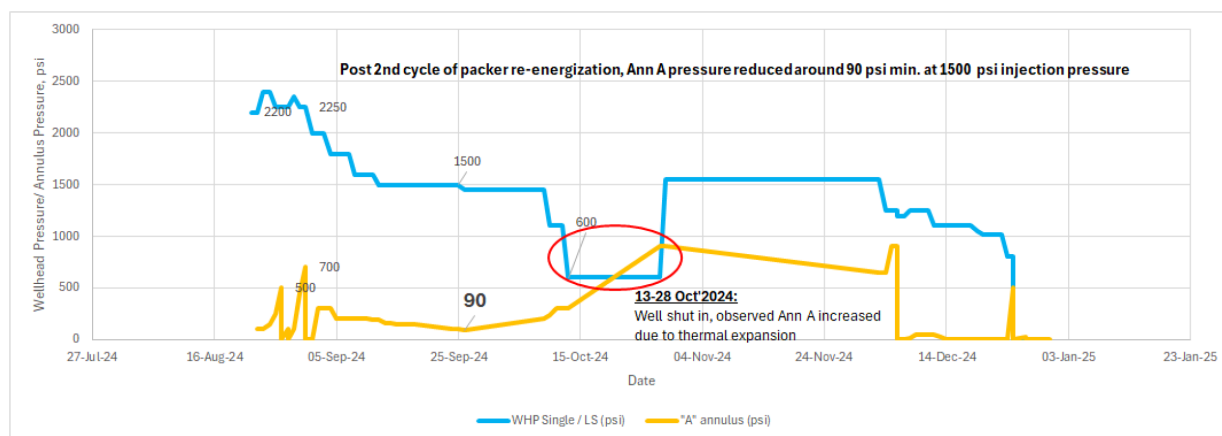


Figure 4—Well "A" - Annulus "A" Bleed down and Build up test Record Post 2nd Cycle Packer Re-energization

In Jan-2025, there was an opportunity to work over the well "A" to cure SCP Annulus A and re-complete the well as SWI equipped with Permanent packer. The well was commissioned in Feb-2025 with 4000 BWPD and 1900 psi wellhead injection pressure. No Annulus A pressure build up has been observed under steady injection mode until to date.

Case 2

Well "B" was drilled and completed in Jan'2024 as SWI with lower completion and equipped with production packer set in the Liner. Well was commissioned in Dec'2024 with 3000 bwpd and 300 psi WHIP. Well had encountered SCP in Annulus A one month after the commissioning phase and thorough SCP investigation was carried out which identified that the source of SCP was the production packer. Annulus A pressure had reached a maximum of 1100 psi (or 28 % of MAASP A) under injection conditions at 900 psi and the pressure was re-built to same value after 24 hours period following the bleed down. Annulus A pressure observed gradually increased to 1370 psi as wellhead injection pressure increased to 1050 psi.

Well was shut in and Annulus A observed built to 2900 psi (73 % of MAASP A) due to thermal expansion effect compounded with SCP. (Refer Fig. 5 Well "B" - Annulus "A" Bleed down and Build up test Record Before Packer Re-energization).

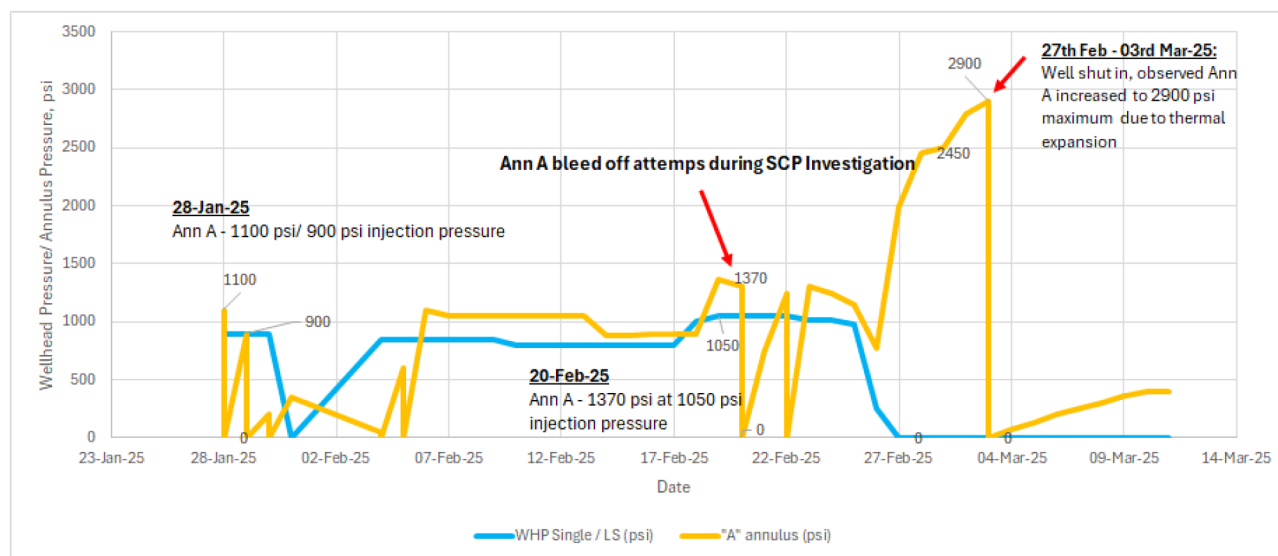


Figure 5—Well "B" – Annulus "A" Bleed down and Build up test Record Before Packer Re-energization

Upon gaining the valuable experience from previous Well "A" packer elastomer re-energization case through rigless solution, similar approach was adopted by pressurizing the tubing against N-test tool to 4500 psi/ 1 hour in stages.

After the re-energization of packer's elastomer, well was resumed injection in staged increments of injection pressure at 500 psi, 750 psi, 1000 psi, 1290 psi and Annulus A pressure was observed at 0 psi under the maximum wellhead injection pressure and no Annulus A PBU observed since Mar 2025 to date under steady injection mode (Refer Fig. 6 Well "B" - Annulus "A" Bleed down and Build up test Record Post Packer Re-energization).

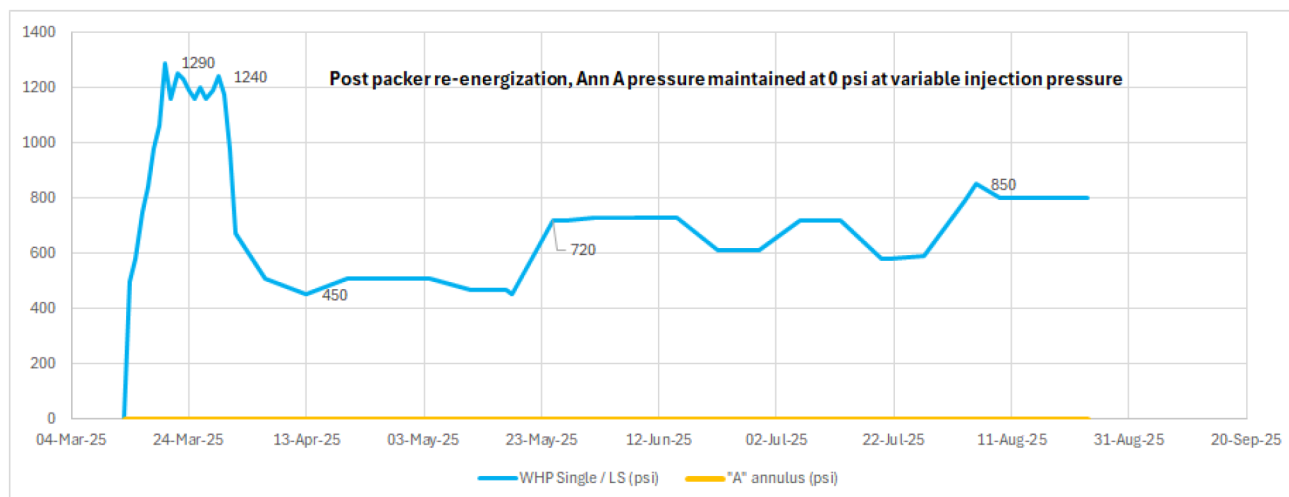


Figure 6—Well "B" - Annulus "A" Bleed down and Build up test Record Post 1st Cycle Packer Re-energization

Table-1 summarizes the SCP investigation steps and results for the above 02 failure cases Well "A" and Well "B":

Table 1—Summarize the SCP Annulus "A" investigation.

Well No	Annulus "A" Bleed down and build up test	Effluent sampling	Wellhead seal test	Tubing Integrity test
Well "A"	Annulus "A" pressure built back to original pressure within 24 hours following the bleed down to zero psi and tracking the Wellhead injection pressure.	Annulus A sample analysis not carried out due to interference	Well head seals satisfactorily holding	Tubing holding 4500 psi/ 1 hour
Well "B"	Annulus "A" pressure built back to original pressure within 24 hours following the bleed down to zero psi and tracking the Wellhead injection pressure	Tubing: 1.12 SG, Ph = 6, H2S = 291 ppm, CO2 = 402 ppm Annulus A: 1.05 SG, Ph = 11.3, H2S = 6 ppm, CO2 = not done due to interference	Well head seals satisfactorily holding	Tubing holding 4500 psi/ 1 hour

Rig-less Curing of SCP Annulus A due to Packer Leak

Following the consultation with Completions team and OEM vendor, the possible cause of packer failure is due to packer being unset or packer elastomers relaxation due to decrease in wellbore temperature during water injection operations. Typically, the packer leak is remediated by rig-based workover which involves significant time and cost and moreover there will be injection down time until the well is recompleted. Other rig-less solution e.g. BISN pumping and Annulus cement plug are still under trial and evaluation phase.

Since there is a high demand of water injection target requirement in the field, other rig-less solutions were discussed with stakeholders and one of the proposals is to carry out the Packer elastomer re-energization operation. This process is basically reenergizing the packer sealing elastomer element by pressurizing the tubing against the conventional down-hole plug set in the tubing to maximum tubing pressure test value, which was applied during completion phase. Batch treatment and wireline package are required to carry out this operation See [Table 2](#) Equipment List required for Packer Re-energization Operations. Tubing clearance will be first conducted to ensure the tubing is clear from any restriction until the nipple profile below the packer and then N-Test tool will be run and set in the nipple profile by Wire line team. Later Batch treatment will be utilized to pump the clean water to fill up the tubing and gradually increase the tubing pressure to target test pressure of 4500 psi in stages and hold for two hours period whilst ensuring Annulus A is full and kept the annulus A open to monitor for any fluid returns.

Table 2—Equipment List required for Packer Elastomer Re-Energization Operations.

Subject	Provision of Wireline/ Slickline Services	Provision of Batch Treatment Package
Scope of work	- Run in hole tubing clearance to locate the Landing Nipple profile - Run in hole N-test tool/ conventional plug and set inside landing nipple profile in the tubing (below the production packer)	- Fill up the tubing capacity above the plug with water - Conduct pressure test in staged increments (Low pressure test and high pressure test) - Packer re-energization by pressurizing tubing up to 4500 psi and hold it for 02 hours.
Dedicated tools	– Wireline unit, N-test tool/ conventional plug, Gauge cutter, etc.	– Batch treatment package including Pumping unit, treating iron, plug valve, check valve, etc.
Operations procedures	– Refer to ADNOC Onshore Well Operations Manual	– Refer to ADNOC Onshore Well Operations Manual
Corresponding parameters	– Pressure, depth, time	– Pressure, time, volume, pumping rate

Upon completion of packer re-energization, well to be resumed injection in stages until maximum injection rate whilst continue monitoring tubing and Annulus A pressure closely. Should there be any build

up pressure in Annulus A during injection period, another attempt of packer elastomer re-energization may be required.

Root Causes Failure Analysis Methodology

Since this failure is related to the same packer type under the same well type (water injector), Packer OEM was requested to investigate the premature packer failure in Well "A" & Well "B". Investigation started with checking the packer material and manufacturing process to ensure the material, tolerances and treatments are correct within the packer design.

The following are the investigation methodology:

- Checking QA/QC documents and record to ensure packer passes through all QA/QC process.
- Checking QA/QC package for packer assembly and testing prior to send to wellsite.
- Checking completion running operation record to check if there is any abnormality observed while running completion string.
- Conduct Tube Move Analyses (TMA) using actual well data to ensure the operating parameters falls within the packer operating envelope.
- Review & discuss the results and findings.

Investigation Findings and Possible Causes

From QA/QC point of view both well's Packer components including slip material were checked and verified by the traceability documents and confirmed to be within the design requirement. Packer QA/QC documents package also reviewed and confirmed that packer was passed through all manufacturing and assembly process. The Final Inspection Sheet and Factory Acceptance Test certificate prior to shipping the packer were checked and all met as per standard. Lastly, Sub assembly make up and pressure testing documents package checked, and all QA/QC checks were done and test charts verified

Table 3—Summarizes the Investigation findings and Possible Causes of Failure on Well "A" and Well "B"

Well Name	Investigation Findings	Possible Causes
Well "A"	Tubing move analysis outcome: • Based on simulation results all loads during injection within packer operating envelope • The tubing length change due to thermal effect (cooling) is -2.85ft which will apply tension force of 65 klbs upward.	• It is possible that the packer element would have been damaged while working on the string to pass the high DLS section with high drag. • It is possible that tubing connection over torqued during rotating the string especially the completion string which consists of different tubing size sections i.e. (4 ½, 3 ½, 4 ½ then again 3 ½").
	Completion Running Operation review • From the job log it was noticed that the completion string faced difficulties to reach Total depth and 10 turns applied on the completion string to sting into the lower completion due to high DLS (dogleg severity).	
Well "B"	Tube Move analysis outcome • TMA Results shows the well operating parameters within the packer operating envelope • TMA results showing the tubing total movement of 3.38ft shrink which generated tension force at the packer of 55 klbs, which is within the packer operating envelope.	• Packer element relaxation, when the packer is first set, the HNBR element is compressed radially against the casing Inside diameter. • Repeated heating/cooling cycles from water injection operations accelerate relaxation. • Elevated downhole temperature softens the elastomer matrix and accelerates chain scission/ rearrangement. • Cooling afterwards can cause micro-gaps if the elastomer shrinks more than the steel housing, leading to partial loss of radial contact to casing inside diameter.
	Completion Running Operation Review • End of well report and Daily Drilling Report was checked and the completion running operation completed smoothly with no reported issue	

Packer Elastomer Re-Energization Principle

The sealing capability of a packer is established during the initial mechanical set, when the elastomer element is compressed radially against the casing ID. Over time, relaxation of the elastomer may occur due to high temperature swing between upper limit and lower limit or due to high axial load, however, the design incorporates a pressure-energization mechanism that restores sealing integrity during subsequent operations.

When tubing pressure is applied, the pressure differential acts on the compressed element stack. This force drives the elastomer radially outward, re-expanding the element into the casing wall. This called Elastomer expansion. The slips remain mechanically locked in position by ratchet rings after the initial set. Pressure does not re-seat the slips since their function is structural anchoring, while pressure re-energization acts solely on the elastomer element. See Fig. 7 - Packer "X" Positional Layout & Fig. 8 - Packer "X" Energization Sequence.

The combination of mechanical pre-set and pressure-assisted re-energization ensures robust sealing integrity throughout the operational envelope (E.g. 4500 psi). This envelope is a representative of expected field stimulation and injection pressure. This level provides a proof test that exceeds typical operating conditions while avoiding overstressing the HNBR Elements or L80 1 % Cr metallurgy components. Pressure test target range 3,500-4,500 psi is widely adopted across OEMs for SIT and FAT (Factory Acceptance Test) protocols, ensuring alignment with standard qualification practices, based on element type and well condition mainly temperature swing between shut in and injection.

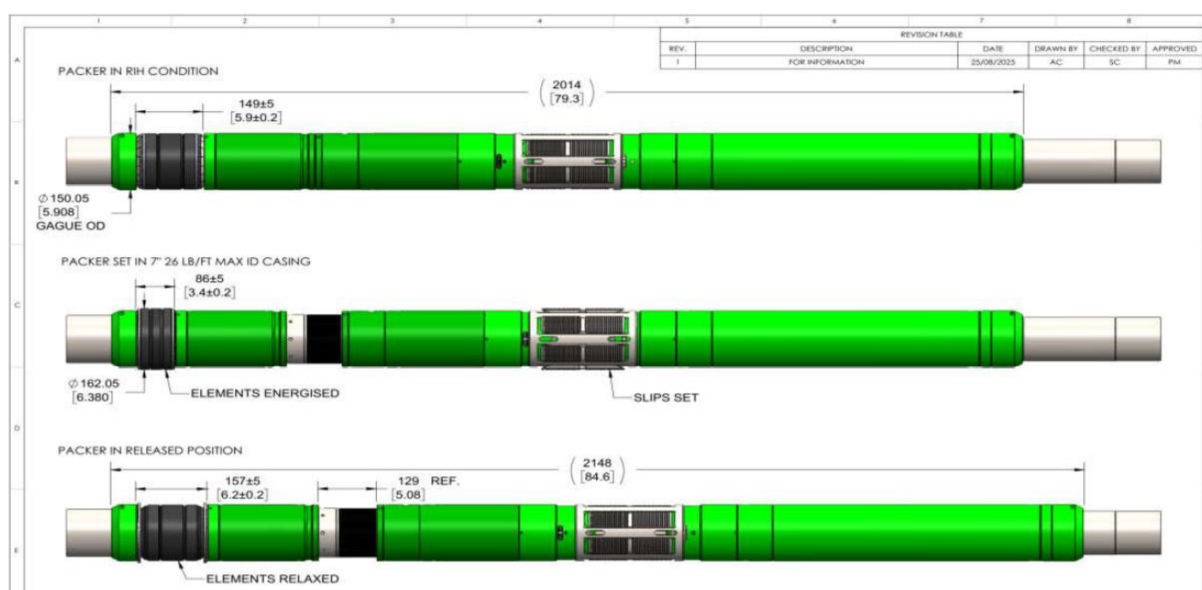


Figure 7—Packer "X" Positional Layout



Figure 8—Packer "X" Energization Sequence

Lesson Learned & Recommendations

Following the success of packer elastomers re-energization operations on Well "A" and Well "B" and detailed root causes failure analysis, best practices and guidelines have been developed along with OEM recommendations in terms of design, installation and operations of Packer to avoid similar issue in the future water injection wells.

Table 4—Summarize of OEM Recommendations over well activity phase

Well phase	Recommendations
Design Phase	- Perform a detailed tubing move analysis (TMA) prior to running the completion string, using the most accurate data available (e.g. actual reservoir pressure, reservoir temperature, surface injection pressure and temperature, and the expected maximum and minimum injection temperatures). - Verify the actual maximum bottom hole temperature (BHT) and the minimum expected injection temperature and confirm whether the current packer temperature rating (45 °C - 150 °C) remains suitable or should be adjusted to a lower temperature rating.
Well Completion Phase	- Ensure clean out the well prior to run completion. - For injector wells it is recommended to set the packer under compression, i.e. set packer prior to fully land the tubing hanger. - Follow the correct packer energization sequence: - Apply 500 psi for 5 minutes to verify plug/barrier disk integrity. - Apply 1,000 psi for 10 minutes to re-confirm plug/barrier disk integrity. - Increase pressure gradually and continuously (without any pauses) to the final energization pressure of 3,500 or 4,500psi and hold for a minimum of 15 minutes. - Increase tubing pressure to the maximum tubing test pressure in line with ADNOC requirement. - Ensure to check the DLS tight spots and simulate the completion string stiffness and consider add more spacer joints to make the string more fixable while passing multiball DLS in short distance. - Minimise completion RIH speed if 5 klbs. and higher noticed while RIH - Avoid rotating the completion string especially at high deviated/ horizontal wells as this might damage the packer element.
Well Commissioning/ Operational Phase	Re energize the packer element in cold conditions if the well temperature dropped significantly from the initial shut in Temp to lower injection temperature.

Conclusions

By restoring the primary barrier integrity using packer re-energization operations through rig-less solution, the well risk category has been significantly reduced from High Medium to Low risk. Also, the company has managed to save approximately +/- \$2.5 million and 15 days on rig-based workover per well while fulfilling the field water injection target of around 2,000 - 3,000 bbls water per day and thus removing the well from Inactive string inventory due to sub-surface integrity issue.

Packer re-energization in water injection wells by pressurizing the tubing with wireline plug set in the downhole riglessly can become one of the alternative solution rather than rig-based workover to restore the primary barrier integrity, thereby meeting the double well barrier envelope policy, where the production packer is identified to be the single source of leak resulting tubing to annulus communication provided no damage caused on the packer element during completion phase and the packer is operated within the operating envelope.

We have applied a similar solution in another newly completed water injection well, where we encountered similar issues related to packer unset and the results were successful on restoring the packer integrity. We also plan to apply for other well types where we encounter similar issues. By adopting this method, deferred injection can be avoided and WO rig slots can be utilized for the acceleration of other projects to achieve positive business impact.

Acknowledgement

The authors would like to express their sincere gratitude to ADNOC Onshore for the opportunity to conduct and present this work. Special thanks are extended to the Integrity & Operations Excellence Department (I&OED) for their continuous support and guidance throughout the project. We acknowledge the valuable contributions of field personnel, OEM technical teams whose insights and operational expertise were instrumental in the successful execution of this study.

We also express our sincere thanks to Mr. Majed Ahmed Alkarbi and Mr. Mohamed Aly Farrag, our direct managers for their leadership and encouragement. Their commitment towards innovation and excellence in well integrity management in ADNOC Onshore has been instrumental behind this innovative solution for restoring packer integrity.

Finally, we appreciate the ADIPEC Technical Committee for providing a platform to share our findings with the wider industry community.

Abbreviation Meaning

ADNOC	Abu Dhabi National Oil Company
API	American Petroleum Institute
BHT	Bottom Hole Temperature
DLS	Dogleg Severity
FAT	Factory Acceptance Test
HNBR	Hydrogenated Nitrile Butadiene Rubber
IOED	Integrity & Operations Excellence Department
MAASP	Maximum Allowable Annulus Surface Pressure
OEM	Original Equipment Manufacturer
POE	Packer Operating Envelope
RIH	Run In Hole
SCP	Sustained Casing Pressure
SIT	System Integrity Testing
SWI	Single Water Injector
TMA	Tubing Move Analysis

WHIP Well Head Injection Pressure
WIMS Well Integrity Management Standard
WO Workover

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